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Salary Efficiency in European Football

To what extent do football players' salaries reflect on-field performance across positions and leagues in Europe's Big Five?

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Academic Year 2025-2026

Executive Summary

This thesis develops a conditional salary efficiency framework applied simultaneously to all five major European football leagues -- the English Premier League, Spanish La Liga, German Bundesliga, Italian Serie A, and French Ligue 1 -- using actual contracted wages, position-specific OLS models with three-way fixed effects (club x league x season), and bootstrap-validated scouting outputs. To the author's knowledge, this combination of scope, positional stratification, and contracted wage data has not appeared in the published literature reviewed for this study. A panel of 6,702 player-season observations spanning 2022-23 to 2024-25 underpins four position-specific wage models that explain between 63% and 71% of log-salary variance. Model residuals are operationalised as conditional salary efficiency scores, measuring how far each player's observed wage deviates from the market benchmark implied by their observable characteristics and competitive context. Four hypotheses are tested. H1 (non-linear wage-performance relationship) receives partial support: the Mincerian age inverted-U is confirmed across all positions, and a significant wage ceiling is found for Midfielders, but superstar wage convexity for Forwards remains directionally plausible without reaching conventional significance. H2 (cross-league variance heterogeneity) is confirmed: Levene's W ranges from 2.44 to 10.96 across position groups, all $p < 0.05$. H3 (club-level efficiency clustering) is confirmed for Forwards (ICC = 0.187) and Goalkeepers (ICC = 0.203) but not for Defenders or Midfielders, consistent with a superstar premium that operates through club prestige in commercially visible positions. H4 (wage rigidity and efficiency persistence) yields the most substantively novel result: players on frozen wages exhibit an estimated AR(1) persistence coefficient of 0.882, compared with 0.328 for players whose wages adjust year-on-year. The interaction coefficient is 0.554 ($t = 17.32$, $p < 10^{-64}$), consistent with contractual rigidity

limiting market correction of initial wage misalignments, though the observational design does not permit a causal interpretation.

An injury-context layer, applied as a post-estimation robustness check, confirms that the top-10 underpaid rankings are stable after adjustment (Jaccard = 1.000), while overpaid rankings show partial sensitivity for players with large documented injury burdens. Applied outputs include a bootstrap-validated shortlist of candidate undervalued players with P10-P90 rank intervals and cross-league transfer routing evidence consistent with institutional wage imprinting from the source league.

Chapter I -- Introduction

1.1 Background and Research Motivation

Player wages are the single largest operating cost for clubs competing in Europe's Big Five leagues, consistently exceeding 50% of total revenues and surpassing 10 billion euros in aggregate per season (European Leagues, 2023). At this scale, the allocation of that wage expenditure across individual players is not merely an accounting question: it is a strategic one. Hall, Szymanski, and Zimbalist (2002) establish Granger causality running from aggregate payroll to competitive outcomes in English football, confirming that wages are consequential competitive inputs. Yet two clubs with identical wage budgets routinely diverge substantially in on-field output, which implies that how those budgets are distributed across individual players carries weight independent of their aggregate level.

Three structural developments make individual wage efficiency particularly timely. First, UEFA's Financial Sustainability Regulations (FSR), introduced in 2022, impose hard annual expenditure constraints under which mispriced multi-year contracts generate compounding regulatory liabilities that cannot be quickly unwound. Gasparini (2024) characterises this as a source of institutional financial risk but provides no individual-level evidence on its magnitude. This thesis contributes evidence on that question. Second, advances in public salary data availability through Capology and FBref now make five-league individual-level efficiency analysis tractable at the required scale. Third, the systematic review by Franceschi et al. (2024), covering 29 empirical valuation studies, confirms that no prior work uses actual contracted wages to construct player-level efficiency scores across all five major European leagues simultaneously -- a specific and actionable gap that this thesis directly addresses.

1.2 Research Problem and Scope

The central argument of this thesis is that any valid salary efficiency measure must condition on all observable sources of legitimate wage variation before attributing residual deviations to genuine mispricing. A Premier League defender earning three times the salary of an equivalent Bundesliga counterpart is not overpaid by construction: that premium reflects institutional wage norms that must be absorbed by league fixed effects before individual pricing deviation can be meaningfully assessed. Position matters equally: goals and assists carry a fundamentally different production-function interpretation for a centre-forward than for a defensive midfielder, and pooling production functions across positional roles introduces systematic bias into any efficiency residual. Both concerns are addressed through league fixed effects and four separate position-specific models.

A further complication, unaddressed in any prior football wage study, is injury-induced playing-time reduction: a player missing extended periods through injury accumulates fewer per-90 statistics, potentially confounding the efficiency score for reasons unrelated to genuine pricing deviation. This thesis addresses this confound explicitly through a post-estimation injury-context robustness layer. The sample covers 6,702 player-season observations across 2022-23 to 2024-25, the longest internally consistent post-pandemic window available and one that spans a stable FSR regulatory environment throughout. The primary research question is: to what extent do football players' salaries reflect their on-field performance once positional roles and competitive contexts are properly taken into account?

1.3 Contributions and Thesis Structure

Relative to the existing literature, this thesis makes four methodological and empirical contributions. First, it applies a player-level salary efficiency framework simultaneously to all five major European leagues using actual contracted wages, a scope not found in the studies

reviewed by Franceschi et al. (2024). Second, it tests the temporal stability of the wage-performance relationship through out-of-sample validation across two complementary holdout configurations, addressing a limitation of Butler et al. (2024). Third, it quantifies rank uncertainty in efficiency-based assessments through 800-resample bootstrap resampling, providing P10-P90 intervals rather than point estimates. Fourth, it introduces an injury-context robustness check to assess whether efficiency scores are systematically confounded by injury-induced playing-time reductions, a potential bias source not previously examined in individual-level football wage studies.

Chapter II reviews the literature and identifies the specific gaps this study addresses. Chapter III develops the theoretical framework and formalises four hypotheses. Chapter IV describes the data and variable construction. Chapter V details the empirical methodology and validation strategy. Chapter VI presents all empirical results. Chapter VII discusses findings, implications, and limitations. Chapter VIII concludes.

Chapter II -- Literature Review

2.1 Football as an Imperfect Labour Market

The foundational insight for any wage efficiency analysis in football is Sloane's (1971) observation that clubs are utility-maximising competitive entities rather than profit-maximising firms in the conventional sense. This objective function allows wages to persistently deviate from marginal productivity, since competitive success can be valued beyond its immediate financial return. The implication is that the football labour market cannot be expected to achieve the efficient wage-productivity alignment that standard competitive labour market models predict. Understanding the sources and magnitude of that misalignment is therefore an empirical rather than a theoretical question.

Hall et al. (2002) establish Granger causality running from aggregate payrolls to competitive performance in English football, confirming that wages are strategic competitive inputs rather than passive reflections of output. Ahtiainen (2018), studying 139 Big Five clubs from 2008-2016, quantifies the stakes: each additional league position generates approximately 1.3 million dollars in incremental revenue through broadcasting distributions and commercial uplift, making wage allocation consequential for financial outcomes. Peeters and Szymanski (2014) document how Financial Fair Play introduced a new layer of constraint on club spending, altering incentives even for clubs with excess financial resources. The FSR architecture, introduced in 2022, intensified those constraints: unlike the earlier framework, FSR evaluates total squad cost annually, meaning an overpaid player's wage compounds as a regulatory liability throughout the contract duration. Franck (2014) demonstrates that the effects of such regulations are enforcement-dependent. This study provides supporting evidence -- significant structural breaks in the wage-performance relationship in Ligue 1 ($F = 7.69$, $p < 10^{-8}$) and Serie A ($F = 4.73$, $p <$

0.001) but not in the Premier League, Bundesliga, or La Liga -- consistent with FSR materially altering wage-setting in leagues furthest from pre-reform financial sustainability.

2.2 Payroll-Performance Relationships: Team-Level Evidence

A large team-level literature confirms a positive payroll-performance relationship across football leagues, but this literature has important limits for the question this thesis poses. Bellavite Pellegrini et al. (2021) and Di Domizio et al. (2022), studying Italian Serie A from 2007-2021, find that relative wages within squads -- not absolute levels -- are the primary predictor of team performance, with balanced internal pay distributions outperforming concentrated ones. Villa and Lozano (2024), applying Data Envelopment Analysis to the Big Five for 2020-2023, find the Bundesliga the second most payroll-efficient league and the Premier League least efficient. This result is directly consistent with this study's finding of compressed Bundesliga and wide Premier League individual efficiency distributions.

The critical limitation of team-level approaches is that they aggregate all players into a single wage figure, making it impossible to identify which individual players drive aggregate inefficiency. Two clubs at the same team-level efficiency score could have entirely different individual mispricing distributions. This limitation motivates the player-level design of this thesis. As Franceschi et al. (2024) confirm, the player-level literature has not yet produced a study that applies efficiency residuals simultaneously across all five major leagues using contracted wages -- the gap this thesis fills.

2.3 Individual Wage Determination

At the individual level, wages consistently increase with goals, assists, playing time, and experience across various football markets (Deutscher and Buschemann, 2016; Berri et al., 2016; Butler et al., 2024). The closest prior individual-level study to this thesis is Butler et al. (2024),

who examine the Premier League and Serie A and find that analytics adoption is associated with wage compression among over- and underpaid players. The present study extends their approach in scope (all five leagues), validation (out-of-sample temporal tests), uncertainty quantification (bootstrap rank intervals), and robustness checking (injury context). Whether these extensions fully resolve the limitations Butler et al. identify is a question addressed in Chapter VII.

Scarfe, Singleton, and Telemo (2021) document a superstar wage premium in MLS where top earners are compensated partly for commercial popularity beyond on-field contribution. The present study finds a pattern consistent with this mechanism at five-league scale and with positional resolution: Club_Wage_Rank_Pct dominates wages for Forwards (coefficient 2.313) and Goalkeepers (1.719) but is statistically indistinguishable from zero for Defenders and Midfielders. This positional asymmetry extends the Scarfe et al. finding with additional institutional specificity. Monteiro (2022) demonstrates that market valuations and contracted wages yield systematically different regression estimates, directly motivating this thesis's exclusive use of contracted wages throughout: market valuations reflect expected future productivity and transfer market sentiment, while contracted wages represent the actual compensation commitment that determines the club's regulatory liability and opportunity cost.

2.4 Performance Measurement and Positional Heterogeneity

A consistent finding across the sports analytics literature is that traditional box-score statistics capture only a fraction of value-added in football. Decroos et al. (2019), formalising the VAEP framework, demonstrate that over 80% of football value-added occurs outside goal contributions -- a finding that directly motivates the inclusion of non-traditional performance metrics in the position-specific specifications used here. Schlenger et al. (2023) systematically validate per-90

normalisation as superior to absolute counts across all 43 position-data-based KPIs they evaluate for cross-player comparability, directly justifying this thesis's per-90 normalisation throughout. The measurement challenge is particularly acute for defensive players. Bradley et al. (2011) demonstrate that raw defensive statistics such as tackles and interceptions primarily reflect team pressing structure rather than individual quality. This system-dependency is confirmed here: within-club defensive statistic variance exceeds between-club variance by 2.4 to 3.3 times, implying that a high tackles-per-90 reading at a high-press club carries less individual quality information than the same reading at a low-press club. This motivates the context-decomposed defensive variables in the Defenders specification. For goalkeepers specifically, Berri et al. (2024) find that wages are driven primarily by team outcomes and co-production effects rather than individual shot-stopping ability, consistent with the dominant Club_Wage_Rank_Pct coefficient found in this study.

Yi (2019) documents systematic technical performance differences across the Big Five leagues, confirming that positional benchmarks are not directly transferable across institutional environments without controlling for league-level structural differences -- the rationale for the league fixed effects and cross-league variance testing in this study.

2.5 Contractual Rigidity and Wage Persistence

The relationship between contract structure and wage persistence has received limited systematic attention in the football economics literature. Theoretical predictions are clear: multi-year contracts insulate wages from within-contract performance revisions, and the longer the contract horizon, the larger the potential accumulated misalignment between wages and current productivity. Mincer (1974) predicts wage arcs over experience that should, in an efficient market, be reflected in forward-looking contract terms; Rosen's (1981) superstar model predicts

convex wage returns to talent in winner-take-all markets. Both predictions generate testable implications for how initial wage commitments track subsequent performance trajectories. Gasparini (2024) identifies the FSR regulatory context as amplifying the cost of contractual rigidity: a club that overpays at signing cannot reverse the commitment within the contract duration, generating annual regulatory exposure throughout. The H4 findings in this thesis provide the first direct quantitative evidence on the persistence implications of that rigidity at individual player level: frozen-wage players exhibit persistence coefficients 2.4 to 2.7 times higher than those on adjusting wages, a gap that directly translates the theoretical regulatory risk into measurable empirical parameters.

2.6 Injuries and Player Value

The relationship between injuries and wages in professional football has not been directly examined at the individual level in the reviewed literature, though Muzaffarov (2025) provides a comprehensive dataset of European football injuries from 2020-2025 that makes empirical investigation feasible. The conceptual concern is specific: if injuries cause playing-time reductions that systematically reduce per-90 performance accumulations, then efficiency scores estimated from performance statistics will conflate genuine wage misalignment with injury-induced statistical suppression. Whether injury risk is rationally priced into long-term contracts at signing -- and therefore should not generate efficiency score adjustments -- is a genuine open question. This thesis addresses the empirical dimension without resolving the theoretical one: the post-estimation injury adjustment layer provides evidence on the robustness of efficiency rankings to injury-burden variation, while acknowledging that the correct interpretation of the adjusted versus unadjusted score depends on assumptions about injury pricing that the design cannot verify.

2.7 Research Gap and Positioning

Taken together, the literature establishes three clear gaps that this thesis addresses. First, the individual-level wage efficiency literature has not extended to all five major European leagues simultaneously using contracted wages (Franceschi et al., 2024). Second, temporal out-of-sample validation of wage models has not been conducted in the football context, leaving the stability of wage-performance relationships untested across seasons (Butler et al., 2024). Third, the individual-level consequences of contractual rigidity for efficiency persistence have not been quantified empirically. This thesis contributes directly to all three gaps.

Chapter III -- Theoretical Framework and Hypotheses

3.1 Wages as Equilibrium Prices in an Imperfect Labour Market

The starting point for the theoretical framework is to treat contracted wages as equilibrium outcomes of bilateral bargaining under institutional constraints, rather than as direct observations of marginal productivity. In a competitive labour market with full information and zero transaction costs, wages would converge to marginal productivity. Football labour markets deviate from this benchmark in three structurally important respects: information asymmetries between clubs and agents about player quality and injury status; monopsonistic elements arising from transfer systems and contract lock-in; and institutional heterogeneity across leagues that generates systematic wage premiums unrelated to individual productivity (Sloane, 1971; Hall et al., 2002).

Under these conditions, the equilibrium wage reflects a combination of individual productivity, position-specific demand characteristics, club financial capacity, agent bargaining power, contract terms negotiated at a point in time, and the institutional wage norms of the league. A valid efficiency measure must separate these components. The conditional efficiency score in this thesis is defined as the residual after all observable institutional and individual characteristics have been absorbed by the fixed effects and performance regressors. It is a benchmark deviation, not a causal estimate of mispricing.

3.2 Position-Specific Productivity and Measurement

A central claim of this thesis is that wage efficiency cannot be estimated without separating the production function by positional role. The argument has two components. The first is empirical: the same observable statistics carry fundamentally different productivity implications across positions. A goal scored by a defensive midfielder and one scored by a centre-forward reflect entirely different relative productivity achievements, because goal-scoring is a high-frequency

primary output for forwards but a low-frequency secondary contribution for midfielders. Pooling these production functions confounds the efficiency residual with positional productivity measurement error.

The second component is about measurement validity. Decroos et al. (2019) demonstrate that the majority of football value-added occurs through non-goal actions, and that positional roles differ systematically in which action types generate value. For defensive players, the system-dependency problem documented by Bradley et al. (2011) means that raw statistics require contextual decomposition before they carry individual quality information. For goalkeepers, co-production with the defence makes individual performance attribution inherently noisy (Berri et al., 2024). These theoretical reasons motivate four separate position-specific models rather than a single pooled specification.

3.3 Conditional Efficiency and Contractual Rigidity

Once the conditional expected wage is estimated, the efficiency score is the gap between what the market would predict given all observable characteristics and what the player actually earns. Two mechanisms can cause this gap to persist over time. The first is information friction: even if clubs wished to correct a misalignment, they may not have timely access to the performance and injury signals that would justify renegotiation. The second is contractual rigidity: multi-year contracts lock wages regardless of performance evolution within the contract period. The FSR regulatory context amplifies both by making the cost of accumulated misalignment visible as a hard annual regulatory liability.

Wage rigidity in the Mincerian tradition arises from implicit contracts that smooth income over the career cycle (Mincer, 1974). In football, this operates through fixed multi-year agreements where the nominal wage does not adjust to within-season performance outcomes. The theoretical

prediction is that players on frozen-wage contracts will exhibit greater persistence of initial efficiency deviations than players on adjusting contracts -- a prediction directly tested in H4.

3.4 Formal Hypotheses

Four hypotheses follow from the theoretical framework and prior literature.

H1 -- Non-linear Wage-Performance Relationship

Mincer (1974) predicts an inverted-U wage arc over the career cycle as returns to experience accumulate then depreciate. Rosen (1981) predicts convex returns to talent in winner-take-all markets for top performers. H1 tests whether the age inverted-U holds across all four position models and whether goal-scoring wage returns display a significant quadratic component -- a ceiling for Midfielders (consistent with positional wage limits) or accelerating returns for Forwards (consistent with superstar convexity). Both positions are tested via an additional NPGls_p90 quadratic specification. Partial support is anticipated: the Mincerian prediction is expected to hold universally, while superstar convexity for Forwards is plausible but less certain given the dominance of Club_Wage_Rank_Pct in that position.

H2 -- Cross-League Variance Heterogeneity

If institutional differences -- in broadcasting revenue distribution formulae, ownership models (notably the Bundesliga's 50+1 rule), competitive balance, and FSR enforcement stringency -- alter the wage-setting process rather than merely shifting its level, then the dispersion of conditional efficiency scores should differ significantly across leagues after fixed effects have absorbed level differences. A wider dispersion in one league indicates that wages deviate more from the observable fundamentals benchmark in that context, regardless of average wage levels. H2 is tested via Levene's equality-of-variance test within each position group across all five leagues simultaneously.

H3 -- Club-Level Efficiency Clustering

If clubs differ systematically in analytical capability, recruitment strategy, or wage-setting philosophy, efficiency scores should cluster within clubs beyond what random assignment would predict. The superstar premium mechanism predicts significant intra-class correlation for Forwards and Goalkeepers -- where Club_Wage_Rank_Pct directly drives wages -- but near-zero ICC for Defenders and Midfielders, where individual fundamentals dominate. H3 is tested via one-way random effects ICC decomposition within clubs by position.

H4 -- Wage Rigidity Amplifies Efficiency Persistence

Multi-year contracts insulate wages from within-contract performance changes, preventing market correction of initial mispricings until contract expiry. H4 tests whether frozen wages (defined as year-on-year real change below 0.5%) generate significantly higher efficiency persistence than adjusting wages, via an OLS interaction of lagged efficiency multiplied by the frozen-wage indicator, controlling for position and season fixed effects. Confirmation translates the theoretical regulatory risk described by Gasparini (2024) into a measurable empirical parameter with direct implications for FSR compliance planning.

Chapter IV -- Data

4.1 Sample Construction and Sources

The analysis uses a pooled cross-sectional panel of player-season observations drawn from Europe's Big Five leagues -- the English Premier League, Spanish La Liga, German Bundesliga, Italian Serie A, and French Ligue 1 -- across three consecutive seasons: 2022-23, 2023-24, and 2024-25. This window is the longest internally consistent post-pandemic period available that is free of pandemic-era wage distortions (salary deferrals and temporary pay reductions) and spans a stable FSR regulatory environment throughout. Performance data were obtained from Stathead (www.stathead.com), the premium interface of FBref that aggregates StatsBomb event-tracking across all five leagues. Salary data were extracted from FBref's wage tables, which draw on Capology estimates of annual contracted base wages. A ScraperAPI middleware layer managed IP rotation for reproducible extraction.

Data cleaning proceeded through five steps. First, all wages were standardised to annual euros using the prevailing exchange rate at the start of each season. Second, player-season records were matched across performance and salary datasets by player name, club, and season; ambiguous matches were resolved through manual verification. Third, players with fewer than 450 minutes of league play were excluded to ensure per-90 statistical stability -- this threshold eliminates squad-depth players whose per-90 metrics would be dominated by small sample noise. Fourth, structural zeros in shooting statistics (players who did not take a single shot in a given season) were handled via indicator dummies rather than exclusion, preserving their wage observations in the panel. Fifth, per-90 performance metrics were winsorised at the 99th percentile to limit the influence of statistical outliers without removing them from the sample.

The final panel contains 6,702 player-season observations distributed across four position groups: Defenders (n = 2,250; 33.6%), Midfielders (n = 3,001; 44.8%), Forwards (n = 925;

13.8%), and Goalkeepers (n = 526; 7.8%). Position classification follows the primary position designation recorded in the FBref source data for the majority of appearances in that season. Players who appeared substantially in multiple positions (more than 30% of minutes in a secondary role) were classified to their primary position and their performance metrics reflect that season's aggregate statistics. Table 4.1 reports descriptive statistics for the full panel.

4.2 Key Variables

The dependent variable throughout is the natural logarithm of the annual contracted base wage in euros. The log transformation is motivated by the strong right skew of the raw wage distribution (mean 2.65 million euros versus median 1.28 million euros) and produces coefficients interpretable as approximate percentage changes. Contracted base wages are used rather than market valuations or total compensation estimates. This choice is deliberate: contracted wages represent the actual financial commitment that determines the club's regulatory liability under FSR, while market valuations reflect anticipated future productivity and transfer market sentiment. Monteiro (2022) demonstrates that the two yield systematically different regression estimates.

Performance metrics are per-90-minute normalised throughout, following Schlenger et al. (2023). The position-specific feature sets are constructed to reflect the distinct production functions of each role. The Defenders specification uses context-decomposed defensive variables -- individual tackles won and interceptions measured as deviations above the club-season mean -- to address Bradley et al.'s (2011) system-dependency finding. For Midfielders, shots on target per 90 and fouls drawn per 90 are included alongside the universal controls. For Forwards, non-penalty goals per 90, conversion rate (goals per shot), offsides per 90 (as a proxy for behind-the-line attacking runs), and Club_Wage_Rank_Pct are included. For Goalkeepers, penalty kick save

percentage and win percentage supplement Club_Wage_Rank_Pct. Table 4.1 provides full variable definitions and expected signs.

The conditional efficiency score is the OLS residual from each position-specific wage model. Positive residuals indicate relative overpayment conditional on all observable characteristics; negative residuals indicate underpayment. The score captures both genuine mispricing and unobservable wage determinants not captured by public statistics -- including leadership premium, image rights contributions, and tactical versatility -- an interpretive caveat maintained throughout.

Frozen-wage indicators are constructed as binary variables equal to one when a player's year-on-year real wage change is below 0.5%, signalling that the contracted base wage did not meaningfully adjust between consecutive seasons. This threshold is calibrated to distinguish fixed multi-year contract years from minor renegotiations. The 0.5% boundary was validated by examining the empirical distribution of year-on-year wage changes, which shows a sharp bimodal structure with a mass near zero (frozen contracts) and a dispersed positive tail (adjusting contracts or new signings).

Table 4.1 -- Variable Definitions and Expected Signs

Variable	Definition	Exp. sign	Source
log_wage	Log annual contracted base wage (euros)	DV	Capology/FBref
Age / Age^2	Age in years; squared term (inverted-U arc)	+/- ***	FBref
Min_pct	Season minutes / maximum possible x 100	+ ***	FBref
NPGls_p90	Non-penalty goals per 90 min (MF, FW, DEF)	+(FW **)	StatsBomb
G_Sh	Goals per shot -- conversion rate (FW)	-(FW **)	Derived
Off_p90	Offsides per 90 min -- attacking runs (FW)	+(FW *)	FBref
SoT_p90	Shots on target per 90 min (MF)	+(MF *)	FBref
Fld_p90	Fouls drawn per 90 min	+/-*	FBref
tkl/int_above_sys	Individual TklW/Int above club-season mean (DEF)	n.s.	Derived
Save_pct_PK	Penalty kick save percentage (GK only)	-(GK *)	FBref
Club_Wage_Rank_Pct	Club rank percentile by total wage bill	+++(FW/GK)	Derived

Note. Stars in 'Exp. sign' reflect confirmed significance in final models (Table 6.3). DV = dependent variable. Full GK specification in Appendix A.

4.3 Injury Data and Matching

Injury records were sourced from Muzaffarov (2025), a publicly available Kaggle dataset covering European football injuries from 2020-2025. Player-season injury records were matched to the main panel through two tiers. Tier 1 (strict) requires exact matching on player name, club, league, and season; Tier 2 (fallback) allows player-league-season matching when the player appears at a unique club in the injury data for that season. Nineteen manual alias mappings were applied to improve strict coverage by 4.9 percentage points. The resulting matched subsample covers 55.8% of the full panel (3,743 of 6,702 observations). Unmatched observations retain the unadjusted baseline efficiency score.

Table 4.2 summarises coverage.

Tier	n	Matched	Rule
Strict (Tier 1)	3,482	100%	Exact: player + club + league + season
Strict + Fallback	3,743	100%	Primary reporting tier; 55.8% of full panel
Full panel	6,702	55.8%	Unmatched rows retain unadjusted score

Table 4.2 -- Injury Context Matching Coverage.

4.4 Data Limitations

Three data limitations bound interpretation and are carried forward to the limitations discussion in Chapter VII. First, salary data capture contracted base wages only: performance bonuses, image rights payments, and signing fees are not included. For commercially active Forwards, image rights income may represent a substantial component of total compensation, implying that the base-wage efficiency score will systematically overstate apparent underpayment for internationally marketed players. Second, the 44.2% unmatched rate in the injury data means that injury-context findings are based on a subsample that may not be fully representative of the overall panel. Third, position classification follows the primary designation in the source data and cannot capture mid-season tactical transitions with precision.

Chapter V -- Methodology

5.1 Research Design

This study employs a quantitative, observational, pooled cross-sectional design with three-way fixed effects. The unit of analysis is the individual player-season observation, preserving the intra-squad variation that is invisible to team-level studies and that constitutes the natural unit of analysis for a labour market where wages are negotiated individually. Strict causal identification is not claimed: wages in professional football result from bilateral bargaining under institutional constraints, not from a clean productivity-to-wage transmission mechanism amenable to experimental design. The appropriate identification exercise is to estimate the conditional expected salary given all observable characteristics and institutional context, then to interpret the residual as conditional pricing deviation rather than a treatment effect.

5.2 Model Specification

The baseline log-wage model estimated separately for each position group p is:

$$log(W_iclt) = alpha + beta_p * X^p_iclt + gamma_Age * Age_it + gamma_Age2 * Age^2_it + theta * MinPct_ict + mu_c + lambda_l + tau_t + epsilon_iclt$$

where W_iclt is the annual contracted wage of player i at club c in league l in season t; X^p_iclt is the position-specific vector of per-90 performance metrics; mu_c, lambda_l, and tau_t are club, league, and season fixed effects respectively; and epsilon_iclt is the idiosyncratic residual that operationalises the conditional salary efficiency score.

The natural log of wages is used as the dependent variable for three reasons. First, the raw wage distribution is strongly right-skewed, violating the normality assumption for OLS residuals on which the efficiency score interpretation rests. Second, log-wage models generate coefficients interpretable as approximate percentage returns, which are economically more meaningful than level effects across a distribution spanning from 20,000 euros to 72 million euros annually.

Third, log transformation reduces the undue leverage of the highest earners on regression coefficients.

Models are estimated separately by position group because production functions differ structurally across roles, as established in Chapter III. Estimating a single pooled model would require the same performance metrics to carry the same wage implications for all positions, a restriction that is both theoretically unjustified and empirically refuted by the significant heterogeneity in coefficient estimates across position groups. LASSO variable selection at the cross-validated optimal regularisation level supports the position-specific feature choices, retaining Club_Wage_Rank_Pct for Forwards and Goalkeepers and excluding it for Defenders and Midfielders.

Club fixed effects absorb time-invariant organisational characteristics that systematically affect wages across all players at that club, including owner wealth, brand prestige, and structural position in the domestic wage hierarchy. League fixed effects absorb structural differences in broadcasting revenues, competitive balance, and regulatory enforcement that generate systematic wage level differentials. Season fixed effects absorb aggregate macroeconomic shocks and industry-wide trends affecting all clubs simultaneously. Together, these three layers ensure the residual reflects idiosyncratic individual pricing deviation rather than structural market variation attributable to observable institutional context.

Standard errors are clustered at the club level, robust to heteroskedasticity and within-cluster correlation. Fixed effects are preferred over random effects because club-level wage norms are correlated with the performance regressors, rendering random effects inconsistent by the Hausman (1978) criterion. Bootstrap-to-HC3 standard error ratios range from 0.74 to 1.38 with

no systematic directional bias, confirming that clustered standard errors are appropriate for these data.

5.3 Validation Strategy

Two temporal out-of-sample exercises test generalisation. The first trains on 2022-24 and evaluates on 2024-25; the second uses the reverse split. These two configurations test forward and backward prediction stability and address the concern that estimated relationships are artefacts of the specific three-season window rather than stable structural features of the wage-performance relationship. A reference OOS R-squared of 0.15 is adopted: models falling below this threshold are considered insufficiently stable for the efficiency residuals to be used as reliable ranking inputs.

Bootstrap resampling uses 800 draws with replacement from the full panel to construct P10-P90 rank intervals for all scouting outputs. This procedure quantifies the sensitivity of individual player rankings to sample composition and provides a natural filter: players whose rank intervals are wide across resamples carry substantially more uncertainty than those with narrow intervals, even if their point-estimate ranks are similar. Jaccard similarity tests across three classification thresholds (plus/minus 0.4, 0.5, and 0.6 standard deviations) confirm that the over- and underpaid classifications are stable to modest changes in the decision boundary.

5.4 Injury Context Layer

The injury-context analysis is applied as a post-estimation robustness check rather than as a primary regressor. Adding injury days directly to the wage model would absorb legitimate injury-risk premia that clubs rationally build into long-term contracts, conflating genuine efficiency signals with risk compensation. Instead, the injury beta (0.056 log-salary units per 30 injury days, estimated by OLS on the matched subsample, standard error approximately 0.011,

statistically significant) is used to construct an adjusted score: the adjusted efficiency score equals the baseline score minus 0.056 times (injury days divided by 30). The injury layer is applied only to matched observations; unmatched rows retain the baseline score with no correction, and all results are reported for both adjusted and unadjusted versions to allow direct comparison.

Chapter VI -- Results

6.1 Descriptive Statistics

Table 6.1 reports descriptive statistics for the full panel. Annual wages are strongly right-skewed (mean 2.65 million euros, median 1.28 million euros), a pattern consistent with the superstar wage structure documented by Scarfe et al. (2021). The efficiency score ranges from -3.060 (Pablo Torre, Barcelona, 2024-25, observed wage 210,000 euros against a model prediction of 4.48 million euros) to +3.998. At the plus/minus 0.5 standard deviation primary threshold, approximately 20% of players are classified overpaid, 20% underpaid, and 60% fairly priced -- near-symmetry confirming no systematic model bias.

Variable	N	Mean	Median	SD	Min	Max
Annual wage (EUR M)	6,702	2.652	1.280	3.887	0.020	72.00
Age (years)	6,702	25.99	26.00	4.28	16	41
Min_pct (%)	6,702	47.1	47.0	26.8	2.6	100
NPGls_p90	6,180	0.113	0.058	0.154	0.000	0.983
Efficiency score	6,702	0.000	0.033	0.652	-3.060	3.998

Table 6.1 -- Descriptive Statistics: Full Panel (n = 6,702 player-seasons, 2022-25). Annual wage in millions of euros. Efficiency score = OLS residual; mean zero by construction. NPGls_p90 covers outfield players (n = 6,180). Source: Capology/FBref (wages); Stathead (performance).

Table 6.2 reports sample composition by position group.

Position	n	Mean wage	Median wage	% Overpaid	% Underpaid	Wage-peak age
Defenders	2,250	EUR 2.48M	EUR 1.25M	18.8%	18.6%	31.5 yrs
Midfielders	3,001	EUR 2.58M	EUR 1.28M	20.5%	19.6%	32.2 yrs
Forwards	925	EUR 3.48M	EUR 1.70M	20.7%	20.3%	32.3 yrs
Goalkeepers	526	EUR 2.27M	EUR 1.20M	21.7%	22.8%	36.0 yrs

*Table 6.2 -- Sample Composition by Position Group. Wage-peak age = -beta_Age / (2 * |beta_Age2|). Forwards command the highest wages, consistent with goal-scoring premiums; Goalkeepers the lowest, consistent with Berri et al. (2024).*

6.2 Model Performance and Temporal Validation

All four models exceed an adjusted R-squared of 0.63, with Defenders reaching 0.713. This level of explanatory power confirms that wages are substantially predictable from observable characteristics and institutional context, though the 29-37% unexplained residual carries systematic structure -- as subsequent results demonstrate. Forwards and Goalkeepers achieve

temporal OOS R-squared of 0.51-0.56 across both holdout directions, confirming strong generalisation for those position groups. Defenders' OOS R-squared of 0.130 marginally misses the 0.15 reference, indicating that Defender efficiency scores carry somewhat more temporal uncertainty than those of other positions. This divergence between high in-sample fit and lower OOS R-squared for Defenders and Midfielders reflects the dominance of time-invariant club fixed effects, which capture stable cross-sectional variance but contribute less to year-on-year within-player prediction.

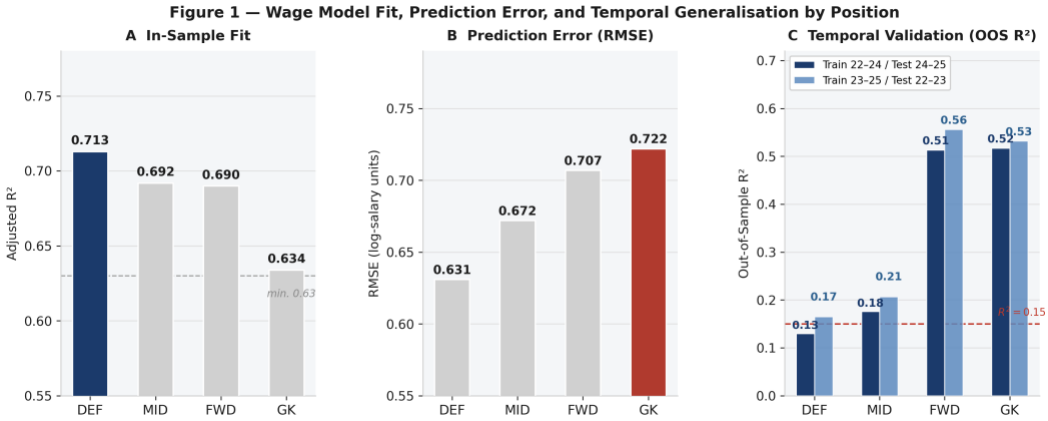

Figure 1. Adjusted R² (A), RMSE in log-salary units (B), and temporal OOS R² from two holdout splits (C) for each position model. Reference line = OOS R² = 0.15.

6.3 Coefficient Estimates

Table 6.3 reports selected coefficient estimates from the four position-specific models. All positions confirm the Mincerian inverted-U wage arc: Age is positive and Age-squared is negative at $p < 0.001$ in all four models. Wage-peak ages differ substantially by position -- Defenders at 31.5 years, Midfielders at 32.2 years, Forwards at 32.3 years, and Goalkeepers at 36.0 years. The 4.5-year gap between Defenders and Goalkeepers is the largest cross-positional difference in the study and reflects the differential depreciation rates of the physical versus cognitive and technical capacities that govern performance at each position. A Goalkeeper

signed at 30 sits near the midpoint of their expected productive career; a Defender signed at 30 is approaching decline.

Min_pct is positive and significant across all four positions, confirming that the manager's deployment decision is the most credible observable quality signal in the wage market -- one that is updated continuously throughout the season based on direct expert observation, making it more credible than performance statistics that can be inflated by tactical system or selection effects.

Variable	Defenders	Midfielders	Forwards	Goalkeepers
Age	0.591 ***	0.565 ***	0.421 ***	0.359 ***
Age^2	-0.009 ***	-0.009 ***	-0.007 ***	-0.005 ***
Min_pct	0.005 ***	0.005 ***	0.009 ***	0.009 ***
NPGls_p90	0.478 n.s.	0.344 n.s.	1.118 **	---
SoT_p90	---	0.137 *	---	---
Fld_p90	-0.072 *	0.050 *	---	---
G_Sh	---	---	-2.767 **	---
Off_p90	---	---	0.161 *	---
Save_pct_PK	---	---	---	-0.003 *
Club_Wage_Rank_Pct	n.s.	n.s.	2.313 ***	1.719 ***
Adj. R^2	0.713	0.692	0.690	0.634
Wage-peak age	31.5 yrs	32.2 yrs	32.3 yrs	36.0 yrs

*Table 6.3 -- Selected Coefficients from Position-Specific Log-Wage Models. *** p<0.001; ** p<0.01; * p<0.05; n.s. = not significant; '---' = not in specification. All models include club, league, and season fixed effects. OLS with club-clustered standard errors. Full tables in Appendix A.*

The most theoretically significant finding is the positional asymmetry in Club_Wage_Rank_Pct. For Forwards (coefficient 2.313, t = 26.3) and Goalkeepers (1.719, t = 13.9), this variable is the dominant predictor, exceeding every individual performance metric in magnitude. For Defenders and Midfielders, it is statistically indistinguishable from zero. This pattern is consistent with a superstar premium mechanism in which wages at commercially prominent clubs incorporate a visibility component not captured by on-field statistics. The interpretation is provisional: the design cannot distinguish between clubs paying above-performance wages because of anticipated

commercial returns, and high-payroll clubs bidding higher across commercially visible positions as a consequence of greater financial resources alone.

The negative G_Sh coefficient for Forwards (-2.767, p = 0.005) is consistent with Decroos et al.'s (2019) VAEP framework: controlling for goals per 90, higher conversion rates imply fewer total shots, and shot-generating actions carry positive expected value even when they do not result in goals. Forwards who score a high proportion of their shots are valued less than those who generate both goals and additional shot volume.

6.4 Efficiency Score Distributions and Cross-League Heterogeneity

Figure 2 presents conditional efficiency score distributions by league and position. Distributions are centred near zero -- wage level differences are absorbed by fixed effects -- but differ substantially in spread, providing primary visual evidence for H2. The Premier League consistently shows the widest dispersion across all four position groups, consistent with Villa and Lozano's (2024) finding of the Premier League as least payroll-efficient at the team level. The Bundesliga exhibits the most compressed distribution, reflecting structural wage discipline from the 50+1 ownership rule. Ligue 1 narrows visibly across the three seasons, consistent with FSR-driven wage compression in that league.

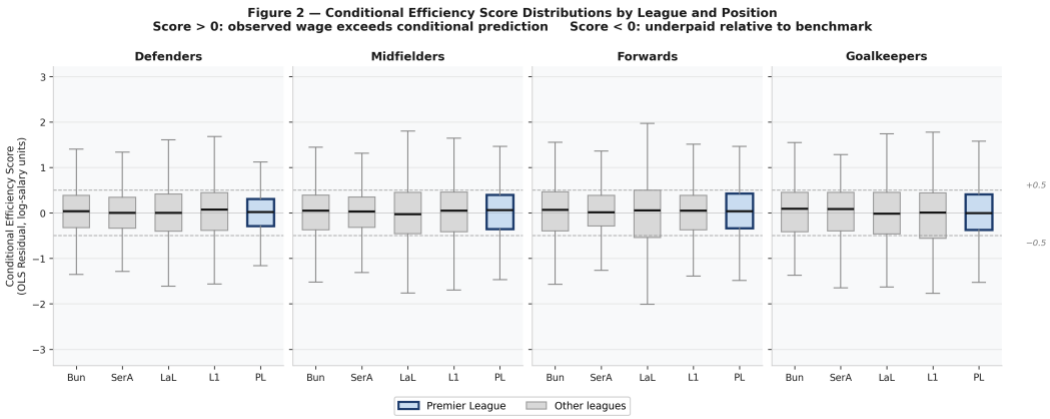

Figure 2. Conditional efficiency score distributions by league and position. Dashed lines mark ±0.5σ classification threshold. H2 confirmed: Levene's W = 2.44–10.96, all p < 0.05 for all four positions.

Figure 3 documents structural wage premiums relative to the Bundesliga baseline. The Premier League commands premiums of 105-512% across positions and seasons -- the upper bound reflecting continued Premier League revenue expansion alongside FSR-driven compression of continental wage floors. The stability of these premiums across the three seasons confirms they are structural institutional features rather than cyclical fluctuations. Without league fixed effects, cross-league comparisons would conflate these institutional differentials with individual pricing deviations.

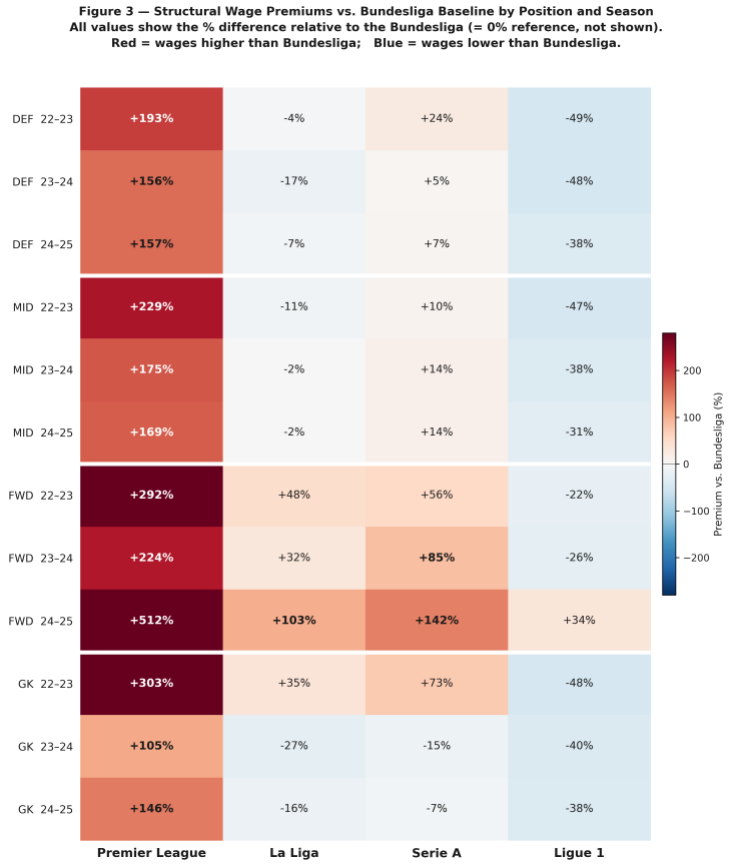

Figure 3. Structural wage premiums (%) vs. Bundesliga baseline by position and season. Bundesliga = 0% reference (not shown); all displayed values are % differences from the Bundesliga for that position-season. Red = wages above Bundesliga; blue = below.

6.5 Hypothesis Test Summary

Hypothesis	Test	Result	Verdict
H1	Age^2; NPGls_p90^2	Age inverted-U: all 4 positions, p<0.001. MID NPGls^2: beta=-1.446, p=0.036. FWD: p=0.261.	Partial
H2	Levene within positions	W = 2.44-10.96, all p < 0.05 for all four positions.	Confirmed

H3	ICC within clubs	FW: 0.187; GK: 0.203; DEF: ~0.000; MID: ~0.000.	Partial
H4	Frozen x lagged efficiency	beta = 0.554, t = 17.32, p < 10^-64; rho_frozen = 0.882; rho_flex = 0.328.	Confirmed

Table 6.4 -- Hypothesis Test Summary. H4 OLS interaction model with position and season fixed effects; Adj. R^2 = 0.520; n = 3,439 consecutive year-pairs.

H1 receives partial support. The Mincerian age inverted-U is replicated across all four positions. The quadratic performance test yields a significant negative coefficient for Midfielders (NPGls_p90-squared = -1.446, t = -2.09, p = 0.036), consistent with a positional wage ceiling at high productivity levels. For Forwards, the positive but insignificant coefficient (p = 0.261) leaves superstar convexity directionally plausible but statistically inconclusive. H2 is confirmed for all four positions. H3 is partially confirmed in the direction predicted by the superstar premium hypothesis: near-zero ICC for Defenders and Midfielders, significant for Forwards and Goalkeepers. H4 is confirmed and is presented in detail in Section 6.6.

6.6 Wage Rigidity and Efficiency Persistence (H4)

Figure 4 presents the H4 findings. Frozen-wage players exhibit estimated AR(1) efficiency persistence of 0.825-0.879 versus 0.359-0.597 for flexible-wage players, a ratio of 1.4 to 2.4 times across positions. Table 6.5 summarises the interaction model results. The interaction coefficient of 0.554 (t = 17.32, p < 10^-64) is the most precisely estimated finding in the thesis. An initial efficiency deviation of 1.5 log-units would be projected to persist at approximately 1.03 log-units by year four of a frozen contract -- a substantially lower correction rate than the 0.053 log-units projected for an adjusting contract over the same period. These projections are based on estimated parameters, not forecasts, and the selection threat noted in Chapter VII applies.

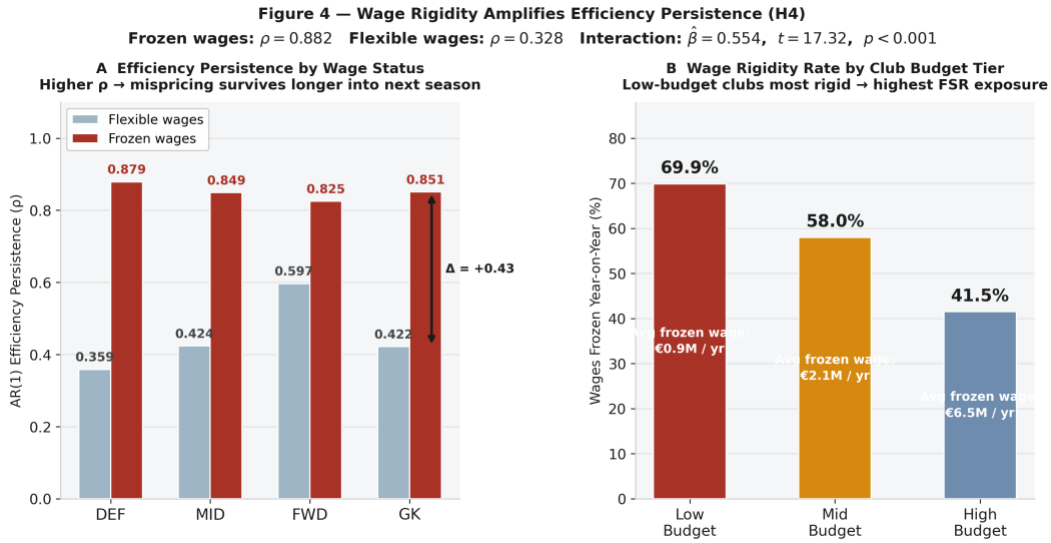

Figure 4. Panel A: AR(1) efficiency persistence by wage status and position. Frozen wages (dark red) persist substantially longer than flexible wages. Panel B: Frozen-wage rate by budget tier. Low-budget clubs — most FSR-vulnerable — are simultaneously the least able to correct accumulated mispricing.

Metric	Flexible wages	Frozen wages
Persistence rho	0.328	0.882
Interaction beta (t-stat)	--- (baseline)	0.554 *** (t = 17.32)
Year-4 carry-over (gap_0 = 1.5)	~0.053 log-units	~1.028 log-units
Frozen rate: Low/Mid/High budget	---	69.9% / 58.0% / 41.5%

*Table 6.5 -- Wage Rigidity and Efficiency Persistence Interaction Model. *** p<0.001. Adj. R^2 = 0.520; n = 3,439 year-pairs. Year-4 carry-over = rho^3 * 1.5. Illustrative projection under estimated persistence parameters; selection caveat applies.*

The budget-tier pattern in Figure 4 (Panel B) is notable: low-budget clubs (below 50 million euros total wage bill) exhibit both the highest frozen-wage rate (69.9%) and the tightest FSR exposure, while high-budget clubs show lower rigidity (41.5%). Whether this asymmetry reflects a structural consequence of the FSR architecture, selection into different contract types, or other confounding factors cannot be determined from this observational design, but the pattern is consistent with the concern that FSR constraints may compound with contractual rigidity in ways that most acutely affect smaller clubs.

6.7 Supplementary Analyses

A. Cross-League Transfer Intelligence

Table 6.6 reports mean arrival efficiency scores by transfer route across 488 observed transfers. Premier League exports arrive overpriced in La Liga (mean arrival score +0.793) because their wages were set under English institutional norms -- a direct consequence of the structural premium differentials documented in Figure 3. Ligue 1 imports to the Premier League arrive underpriced (-0.153) because French wage floors are structurally lower. These patterns are consistent with wage floors following players across institutional environments, a finding that has implications for how cross-league negotiations are best approached.

Transfer route	n	Mean arrival eff.	Interpretation
Ligue 1 to Premier League	20	-0.153	Arrives underpriced
La Liga to Premier League	20	-0.050	Near-fairly priced
Bundesliga to Premier League	19	+0.256	Moderate premium
Premier League to La Liga	23	+0.793	Substantially overpriced
Premier League to Ligue 1	12	+0.704	Substantially overpriced
Serie A to La Liga	10	+0.573	Overpriced

Table 6.6 -- Mean Arrival Efficiency Score by Transfer Route (488 transfers). Arrival score = player's conditional efficiency score in first season at new club vs. destination-league benchmark.

B. Nationality Premium Analysis

ANOVA on efficiency scores by nationality yields $F = 9.21$ for Defenders, $F = 8.90$ for Midfielders, and $F = 12.39$ for Forwards (all $p < 0.001$); Goalkeepers are not significant ($F = 0.86$). The most underpaid nationality groups are Algerian (mean efficiency -0.345), Cameroonian (-0.154), and Polish (-0.109); the most overpaid are Belgian (+0.289) and Ukrainian (+0.233). These patterns are consistent with a combination of agent network concentration, fan demand differentials, and supply-demand imbalances in specific nationality segments of the transfer market. The appropriate interpretation within this design is that raw efficiency scores partially reflect market-wide nationality patterns in addition to individual pricing deviation; treating nationality effects as potential correction layers rather than evidence of discrimination is the more defensible inference.

C. Bootstrap-Validated Scouting Rankings

Figure 5 presents the 15 most underpaid players by bootstrap mean rank. Pablo Torre (Barcelona, 2024-25) is the maximum of the underpaid distribution: observed wage 210,000 euros against a model prediction of 4.48 million euros, efficiency score -3.060, rank 1 in every resample in which he is drawn (bootstrap interval [1, 1]), and a 63.5% appear rate reflecting the standard bootstrap property of approximately 36.8% omission probability per resample. The interval [1, 1] and the 63.5% appear rate measure different properties of ranking stability and are not in tension. His case survives all robustness checks: the injury adjustment is only 0.021 log-units against a 3.060-unit gap; Spanish nationality generates near-zero expected nationality correction; and his rank is invariant across the 800 resamples in which he appears.

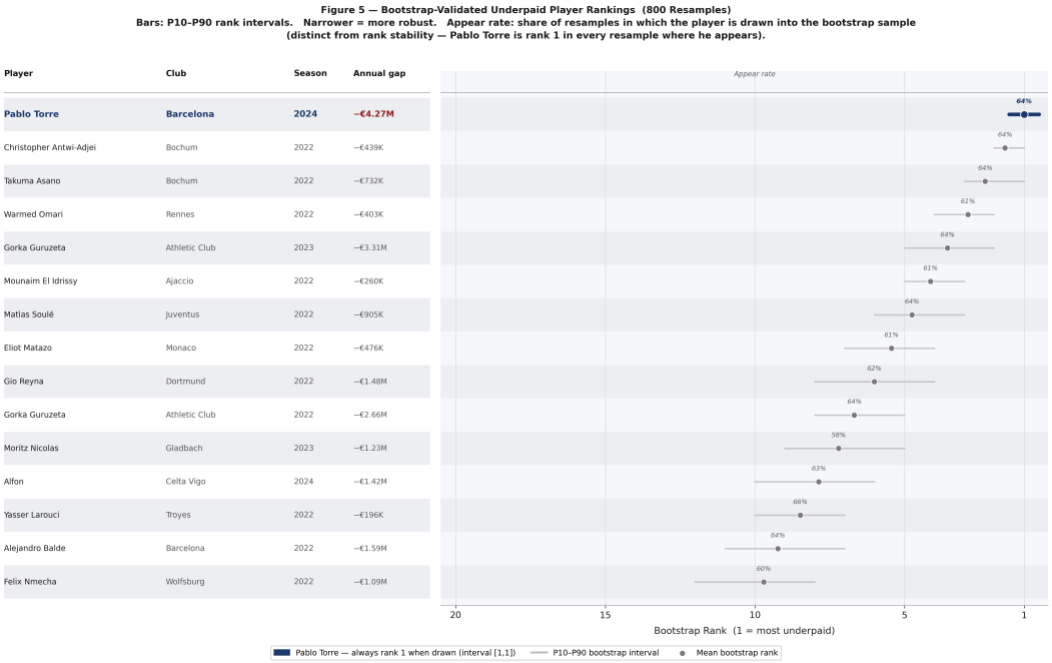

Figure 5. Bootstrap-validated underpaid rankings (800 resamples). Left: player, club, season, and annual salary gap (predicted minus observed wage) for all 15 players. Right: P10–P90 rank intervals and appear rates. Narrower interval = more robust ranking.

6.8 Injury Context Analysis

The estimated injury beta is 0.056 log-salary units per 30 injury days -- statistically significant but economically modest. A player missing 90 days receives a 0.168-unit adjustment, approximately 26% of the efficiency score standard deviation. The mean injury burden for

matched observations is 59.9 days and the median is 40.0 days. The asymmetric result is the key finding of this analysis. The top-10 underpaid ranking is identical before and after injury adjustment (Jaccard = 1.000), confirming that underpaid signals in the top-10 do not reflect injury-induced statistical suppression. For the overpaid rankings, adjustment is more consequential: Thibaut Courtois (258 injury days, moves five positions) and Manuel Neuer (281 days, moves two positions) shift out of the top-20 after adjustment.

The Courtois case illustrates the practical value of the injury layer: 258 documented injury days in 2023-24 generate a substantial downward correction to the baseline efficiency score, partially reversing the apparent overpayment signal. Whether the adjusted or unadjusted score better represents genuine wage misalignment depends on assumptions about whether injury risk is rationally priced into long-term contracts at signing -- a question the design does not resolve. The operational implication is asymmetric: underpaid signals in the top-10, which are stable after adjustment, can be used as screening inputs without an injury caveat; overpaid signals at any rank should be accompanied by explicit injury history verification before operational use.

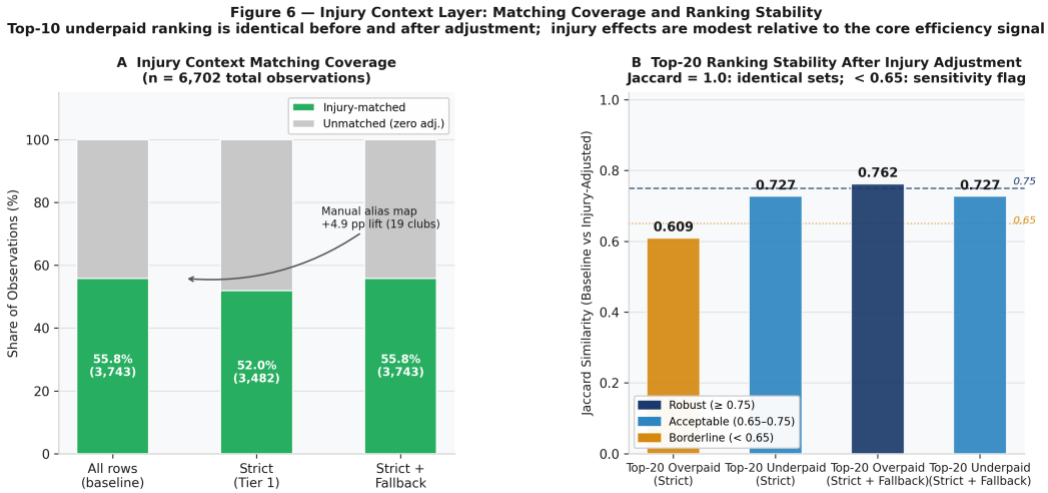

Figure 6. Panel A: Injury matching coverage (55.8% of full panel matched). Panel B: Jaccard similarity before and after adjustment. Top-10 underpaid: Jaccard = 1.000 (perfectly stable). Overpaid top-20: Jaccard = 0.762 (partial sensitivity driven by injury-heavy players).

Chapter VII -- Discussion, Implications, and Limitations

7.1 Synthesis and Interpretation

Six principal findings characterise the football labour market as institutionally differentiated and imperfectly but not randomly efficient. First, wages are substantially predictable from observable characteristics: 63-71% of log-salary variance is explained across position groups, confirming the market is not arbitrary. The 29-37% unexplained residual is not noise -- it carries systematic structure across leagues, positions, budget tiers, and regulatory regimes, confirming genuine pricing deviations that the framework can detect.

Second, the wage market is positionally segmented between two fundamentally different sub-markets. Defenders and Midfielders are priced close to individual fundamentals (ICC approximately 0.000), while Forwards and Goalkeepers operate under a superstar premium where club wealth drives wages directly above the performance benchmark (ICC approximately 0.19-0.20). This segmentation -- corroborated simultaneously by the Club_Wage_Rank_Pct positional coefficient asymmetry and the ICC results -- extends the superstar premium pattern that Scarfe et al. (2021) documented in MLS to five European leagues, with additional positional resolution not present in prior five-league work. The practical implication is that efficiency scores carry different levels of credibility by position: for Defenders and Midfielders, efficiency scores represent cleaner signals of individual pricing deviation; for Forwards and Goalkeepers, part of any negative score may reflect an underestimated image rights contribution that the base-wage model cannot capture.

Third, the H2 confirmation indicates that cross-league wage-setting processes differ in dispersion as well as level. The transfer routing patterns in Table 6.6 are consistent with this finding: players moving from the Premier League to La Liga arrive with systematically positive efficiency scores relative to the Spanish benchmark, reflecting wages set under a structurally

higher wage floor. This pattern is stable across 488 observed transfers and is unlikely to be explained by idiosyncratic individual quality differences alone. The most parsimonious interpretation is institutional wage imprinting from the source league, though the design cannot rule out selection on unobservable player characteristics.

Fourth, the estimated association between wage rigidity and efficiency persistence is the most substantively novel result. The interaction coefficient of 0.554 ($t = 17.32$) indicates that frozen wages are associated with substantially stronger year-on-year persistence of efficiency deviations. This is consistent with the prediction that contractual rigidity limits within-contract wage correction. The selection threat -- that players on fixed versus adjusting contracts may differ on unobservable characteristics correlated with both contract type and performance trajectory -- cannot be excluded from this observational design and qualifies any causal reading of the persistence parameters.

Fifth, the injury robustness check produces an asymmetric result with clear operational implications: top-10 underpaid rankings are stable after adjustment (Jaccard = 1.000), while overpaid rankings show partial sensitivity driven by players with large documented injury burdens. Sixth, nationality is an empirically significant correlate of efficiency scores for three of four position groups, and treating these patterns as correction layers rather than as evidence of pure individual mispricing is the appropriate interpretation within this design.

Two results are only partially confirmed. H1's superstar convexity prediction for Forwards is directionally supported but does not reach conventional significance ($p = 0.261$), plausibly because Club_Wage_Rank_Pct already absorbs most of the variance that superstar effects would otherwise explain. H3's club-level clustering prediction is confirmed for positions where the superstar premium mechanism operates but not for positions where it does not -- a partial

confirmation that is itself theoretically informative, as it demonstrates that the mechanism is position-specific rather than general.

7.2 Practical Implications

The empirical findings suggest several directions that practitioners might explore, subject to the data and design limitations identified in Section 7.3. These observations are appropriately treated as illustrative signals rather than operational prescriptions: efficiency scores are OLS residuals from observational models, and the gap between a conditional wage benchmark and a player's contracted salary does not by itself constitute evidence of a negotiable undervaluation.

For contract structure, the H4 findings are consistent with the view that multi-year fixed-wage commitments are associated with greater persistence of initial pricing deviations than shorter or performance-linked structures. For positions where Club_Wage_Rank_Pct is large (Forwards and Goalkeepers), this may be particularly consequential given that wages in those positions already incorporate a prestige component that is not performance-contingent. Whether more flexible contract structures would reduce efficiency persistence is a causal question the observational design cannot answer.

A structured screening approach suggested by the findings would apply three sequential filters. First, account for nationality-level patterns: players from nationalities with documented market-wide premia have efficiency scores that partly reflect those patterns rather than purely individual pricing deviation. Second, apply injury verification for overpaid signals before operational use. Third, prioritise candidates with narrow bootstrap intervals and appear rates above 60%, as these represent rankings stable to data perturbation. The cross-league transfer routing evidence suggests that benchmarking against destination-league norms rather than source-league contracted wages would produce more accurate assessments in cross-league negotiations.

The budget-tier pattern raises a question about whether the current FSR architecture creates incentives consistent with its stated objectives -- specifically, whether hard spending constraints at lower-budget clubs compound with contractual rigidity in ways that limit flexibility to correct accumulated mispricing. This observation is tentative, rests on secondary analysis of the efficiency data, and evaluating FSR regulatory design is outside the scope of this thesis.

7.3 Limitations

Six limitations bound the interpretation of findings from this thesis. They are presented not as apologies but as the boundaries within which the evidence is valid, following the principle that acknowledged limitations strengthen rather than weaken a study.

First, salary data capture contracted base wages only. Total compensation -- including performance bonuses, image rights payments, and signing fees -- is unmeasured. For commercially active Forwards, image rights income may represent a substantial component of true compensation, implying that base-wage efficiency scores will systematically overstate apparent underpayment for internationally marketed players. This limitation is most consequential for the top of the underpaid distribution, where high-profile young players on conservative base wages may have significant unobserved total compensation.

Second, the panel is observational. Coefficients are conditional associations, not causal effects. The multi-year contract structure mitigates the primary simultaneity concern between wages and playing time -- wages are set at signing and are not renegotiated within a season based on performance fluctuations -- but does not eliminate all endogeneity concerns. Selection into wage levels based on unobservable player characteristics, agent quality, or club negotiating strength cannot be ruled out. The efficiency residuals should be interpreted as conditional benchmarks rather than causal mispricing estimates.

Third, the Forwards model condition number (4.4 x 10^19) indicates near-perfect collinearity among some predictor interactions. LASSO confirms the stability of the four significant coefficients under penalisation, but individual Forwards coefficient estimates carry more specification uncertainty than those of other positions. This means that the Forwards efficiency scores are somewhat more sensitive to model specification choices than those of Defenders and Midfielders.

Fourth, the 44.2% unmatched rate in the injury data means that injury-context findings are based on a subsample that may not fully represent the overall panel. Systematic differences between matched and unmatched observations -- for instance, if injury records are more complete for players at larger or more administratively organised clubs -- could bias the injury beta estimate in ways the design cannot detect.

Fifth, several potentially important wage determinants are not captured by publicly available statistics: tactical versatility, leadership and dressing-room influence, market pressures at specific positions, agent relationships, and club-specific salary band policies. To the extent these factors are correlated with position, league, or performance metrics, they introduce bias into the efficiency residual in directions that cannot be fully characterised.

Sixth, external validity is constrained by the specific institutional context examined (the Big Five, 2022-25, post-FSR introduction). The efficiency parameters estimated here reflect a particular regulatory regime and a particular phase of the transfer market. Results may not generalise to lower-tier leagues, non-European markets, or periods characterised by materially different regulatory frameworks.

7.4 Directions for Future Research

Three research extensions follow directly from the limitations. First, incorporating total compensation data -- image rights and performance bonuses increasingly disclosed in Spanish and French regulatory filings -- would resolve the base-wage limitation and likely reveal larger misallocations for high-profile Forwards. Second, implementing VAEP-based performance measurement for Defenders and defensive Midfielders would produce individual action values that already condition on team system context, potentially improving the Defenders model's OOS R-squared by replacing system-contaminated raw statistics with genuinely individual quality measures. Third, a difference-in-differences design exploiting the 2022 FSR introduction -- with Ligue 1 and Serie A (which show significant Chow structural breaks) as treated leagues and the Premier League and Bundesliga as controls -- would provide causal identification of FSR's effect on individual wage-setting efficiency, moving beyond the observational persistence evidence presented here.

Chapter VIII -- Conclusion

The football labour market exhibits systematic patterns that a conditional efficiency model can detect and quantify. Wages respond to age, playing time, and performance with sufficient regularity across all five major European leagues for the framework to produce meaningful efficiency residuals. The four position-specific models explain 63-71% of log-salary variance and generalise out of sample with OOS R-squared of 0.51-0.56 for Forwards and Goalkeepers -- the range where the framework is most reliable as a practical screening tool.

The primary research question -- to what extent do football players' salaries reflect on-field performance once positional roles and competitive contexts are properly taken into account -- receives a nuanced answer. Wages are substantially aligned with observable performance and institutional context for the large majority of players: roughly 60% fall within plus/minus 0.5 standard deviations of the conditional benchmark. The remaining 40% exhibit deviations that are not random. They are structured by position (a superstar premium for Forwards and Goalkeepers not present for Defenders and Midfielders), by league (wider dispersion in the Premier League, more compressed in the Bundesliga), by regulatory context (significant structural breaks in Ligue 1 and Serie A coinciding with FSR introduction), and by contract type (frozen wages generating persistence ratios two to three times higher than adjusting wages).

The estimated persistence association (rho = 0.882 for frozen wages) is the most substantively novel finding. It translates the theoretical regulatory risk identified by Gasparini (2024) into a measurable empirical parameter: an initial efficiency deviation of 1.5 log-units is projected to persist at approximately 1.03 log-units by year four of a frozen contract, compared with approximately 0.05 log-units for an adjusting contract. The selection caveat applies, but the estimated gap is large enough to remain meaningful even under moderate violations of the identifying assumption.

Pablo Torre earning 210,000 euros against a model prediction of 4.48 million euros is the most vivid illustration in the data of contract-timing misalignment. His gap survives injury correction (0.021-unit adjustment against a 3.060-unit deviation), nationality correction (Spanish players average near-fair efficiency scores), and bootstrap instability testing (rank 1 in every resample in which he is drawn). The appropriate inference is not that his club has made an irrational decision -- the model cannot account for the full terms of his original contract, any image rights arrangements, or tactical considerations not captured by public statistics. The inference is that his observed base wage falls substantially below the conditional market benchmark, and that this gap is stable to all robustness checks applied.

What the framework provides is a disciplined, reproducible benchmark against which individual wage commitments can be assessed, with quantified uncertainty on all practical outputs. The conditional efficiency score is not a definitive statement of true player worth. It is a systematic first filter -- one grounded in market norms rather than subjective judgment, robust to the institutional and positional heterogeneities that make naive wage comparisons misleading, and honest about the boundaries of what observational evidence can establish.

References

Ahtiainen, S. (2018). Top 5 European football leagues [Master's thesis, Aalto University School of Business].

Bellavite Pellegrini, C., Caruso, R., & Di Domizio, M. (2021). Relative wages, payroll structure and performance in soccer. Quaderno n. 15, Universita Cattolica del Sacro Cuore.

Berri, D., Buraimo, B., Rossi, G., & Simmons, R. (2016). Pay and performance in Italian football. Birkbeck Sport Business Centre Research Paper Series, 9(4).

Berri, D., Butler, D., Rossi, G., Simmons, R., & Tordoff, C. (2024). Salary determination in professional football: Empirical evidence from goalkeepers. European Sport Management Quarterly, 24(3), 624-640.

Bradley, P. S., Mohr, M., Bangsbo, J., & Krustrup, P. (2011). High-intensity running in English FA Premier League soccer matches. Journal of Sports Sciences, 29(2), 117-126.

Butler, D., Farnell, A., & Simmons, R. (2024). Do sports analytics affect footballer pay? Frontiers in Behavioral Economics, 3, Article 1490871.

Decroos, T., Bransen, L., Van Haaren, J., & Davis, J. (2019). Actions speak louder than goals: Valuing player actions in soccer. arXiv:1802.07127.

Deutscher, C., & Buschemann, A. (2016). Does performance consistency pay off financially for players? Journal of Sports Economics, 17(1), 27-43.

Di Domizio, M., Bellavite Pellegrini, C., & Caruso, R. (2022). Payroll dispersion and performance in soccer. Contemporary Economic Policy, 40(3), 513-525.

European Leagues. (2023). The financial landscape of European football. European Leagues Organisation.

Franceschi, M., Brocard, J.-F., Follert, F., & Gouguet, J.-J. (2024). Determinants of football players' valuation: A systematic review. Journal of Economic Surveys, 38(3), 577-600.

Franck, E. (2014). Financial fair play in European club football. International Journal of Sport Finance, 9(3), 193-217.

Gasparini, S. (2024). Innovative financial strategies in football [Master's thesis, Ca' Foscari University of Venice].

Hall, S., Szymanski, S., & Zimbalist, A. S. (2002). Testing causality between team performance and payroll. Journal of Sports Economics, 3(2), 149-168.

Hausman, J. A. (1978). Specification tests in econometrics. Econometrica, 46(6), 1251-1271.

Malikov, D., Jung, P., & Kim, J. (2025). Predicting soccer player salaries. Applied Sciences, 15(15).

Mincer, J. (1974). Schooling, experience, and earnings. Columbia University Press.

Monteiro, J. P. F. (2022). Performance and actual pay in the top 5 European football leagues [Master's thesis, NOVA IMS].

Muzaffarov, S. (2025). European football injuries 2020-2025 [Data set]. Kaggle.

Peeters, T., & Szymanski, S. (2014). Financial fair play in European football. Economic Policy, 29(78), 343-390.

Rosen, S. (1981). The economics of superstars. American Economic Review, 71(5), 845-858.

Scarfe, R., Singleton, C., & Telemo, P. (2021). Extreme wages, performance, and superstars in a market for footballers. Industrial Relations, 60(1), 84-118.

Schlenger, J., Wunderlich, F., Raabe, D., & Memmert, D. (2023). Systematic analysis of position-data-based KPIs in soccer. International Journal of Computer Science in Sport, 22(1), 1-22.

Sloane, P. J. (1971). The economics of professional football. Scottish Journal of Political Economy, 18(2), 121-146.

Villa, G., & Lozano, S. (2024). A payroll efficiency analysis of Europe's top football leagues. IMA Journal of Management Mathematics.

Yi, Q. (2019). Differences in technical performance of players from 'The Big Five'. Frontiers in Psychology, 10, Article 2738.

Appendix A -- Full Coefficient Tables

All models include club, league, and season fixed effects. Standard errors clustered at club level.

*** p<0.001; ** p<0.01; * p<0.05.

Table A1 -- Defenders (n = 2,250; Adj. R^2 = 0.713; RMSE = 0.631; 95 clusters)

Variable	Coef.	SE	t	CI 2.5%	CI 97.5%
Age ***	0.591	0.042	14.23	0.509	0.672
Age^2 ***	-0.009	0.001	-12.44	-0.011	-0.008
Min_pct ***	0.005	0.001	7.92	0.004	0.006
team_press_intensity	0.124	0.126	0.99	-0.123	0.372
tkl_above_system	-0.026	0.035	-0.75	-0.095	0.042
int_above_system	0.018	0.036	0.50	-0.052	0.088
NPGls_p90	0.478	0.474	1.01	-0.451	1.407
Fld_p90 *	-0.072	0.032	-2.28	-0.134	-0.010

Table A2 -- Midfielders (n = 3,001; Adj. R^2 = 0.692; RMSE = 0.672; 95 clusters)

Variable	Coef.	SE	t	CI 2.5%	CI 97.5%
Age ***	0.565	0.035	16.25	0.497	0.633
Age^2 ***	-0.009	0.001	-13.71	-0.010	-0.008
Min_pct ***	0.005	0.001	9.56	0.004	0.006
SoT_p90 *	0.137	0.062	2.20	0.015	0.260
Fld_p90 *	0.050	0.021	2.35	0.008	0.092
NPGls_p90	0.344	0.224	1.54	-0.095	0.784
G_Sh	-0.431	0.236	-1.83	-0.894	0.032

Table A3 -- Forwards (n = 925; Adj. R^2 = 0.690; RMSE = 0.707; 96 clusters)

Variable	Coef.	SE	t	CI 2.5%	CI 97.5%
Age ***	0.421	0.059	7.18	0.306	0.537
Age^2 ***	-0.007	0.001	-6.11	-0.009	-0.004
Min_pct ***	0.009	0.001	8.62	0.007	0.011
NPGls_p90 **	1.118	0.405	2.76	0.323	1.913
G_Sh **	-2.767	0.979	-2.83	-4.688	-0.847
Off_p90 *	0.161	0.067	2.40	0.029	0.294
Club_Wage_Rank_Pct ***	2.313	0.088	26.25	2.140	2.486

Table A4 -- Goalkeepers (n = 526; Adj. R^2 = 0.634; RMSE = 0.722; 58 clusters)

Variable	Coef.	SE	t	CI 2.5%	CI 97.5%
Age ***	0.359	0.075	4.76	0.211	0.507
Age^2 ***	-0.005	0.001	-3.90	-0.007	-0.002
Min_pct ***	0.009	0.001	7.77	0.007	0.012
Save_pct_PK *	-0.003	0.002	-1.99	-0.006	0.000
W_pct	0.419	0.242	1.73	-0.056	0.894
Club_Wage_Rank_Pct ***	1.719	0.124	13.85	1.475	1.963

-- End of Thesis --